

The rotational and magnetic properties of Polaris: Challenges to standard stellar models

ApJ In Press, arxiv:2603.09876

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Image Credit: J.-C. Cuillandre, CFHT



Queen's
UNIVERSITY

Introduction: Polaris the North Star

Distance

133 pc

System Components

Polaris (Aa + Ab) + B

Spectral Types

(F8Ib + F6V) + F3V

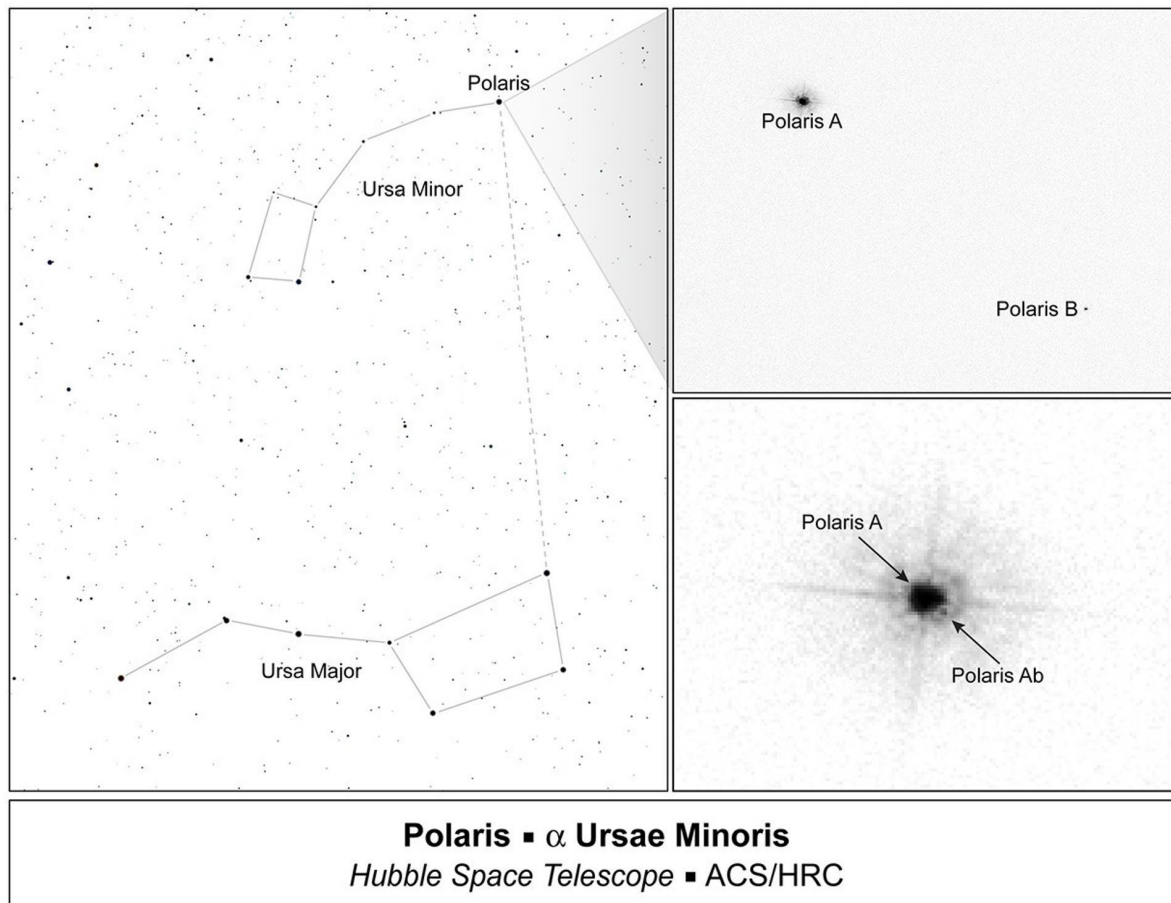
Inner Orbit (Evans+ 2024)

$P_{\text{orb}} = 29.4$ years

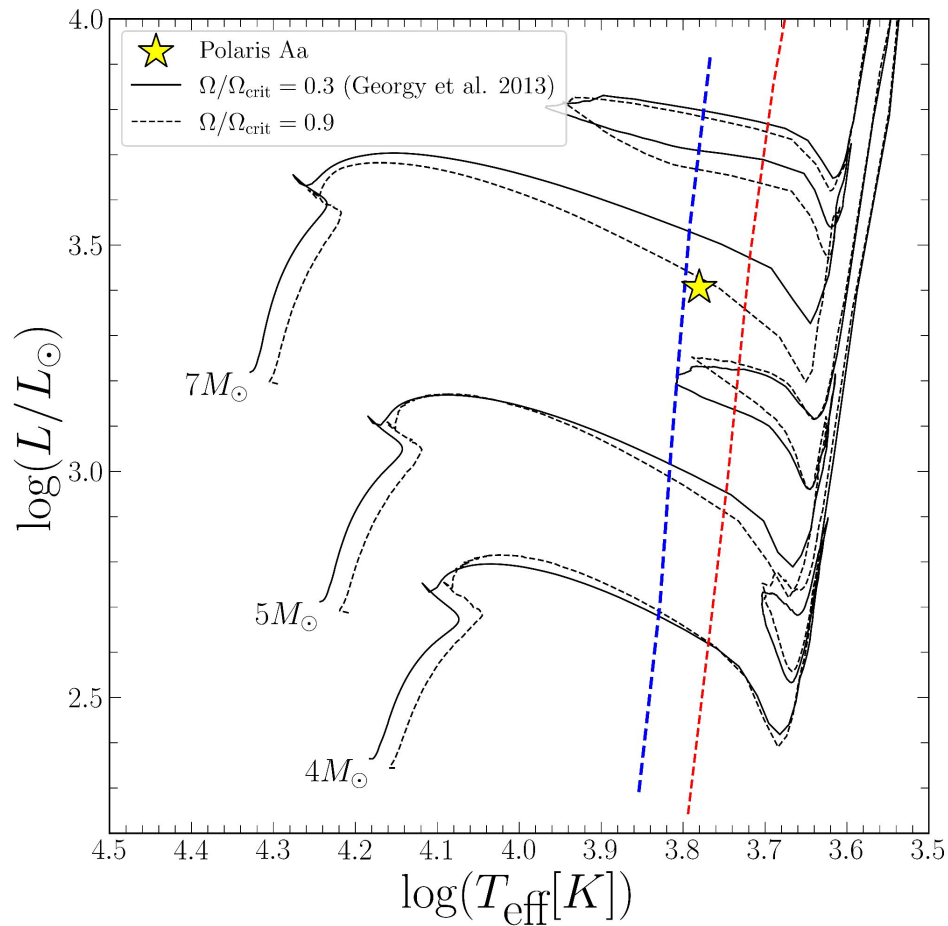
$e = 0.62$

Dynamical Mass

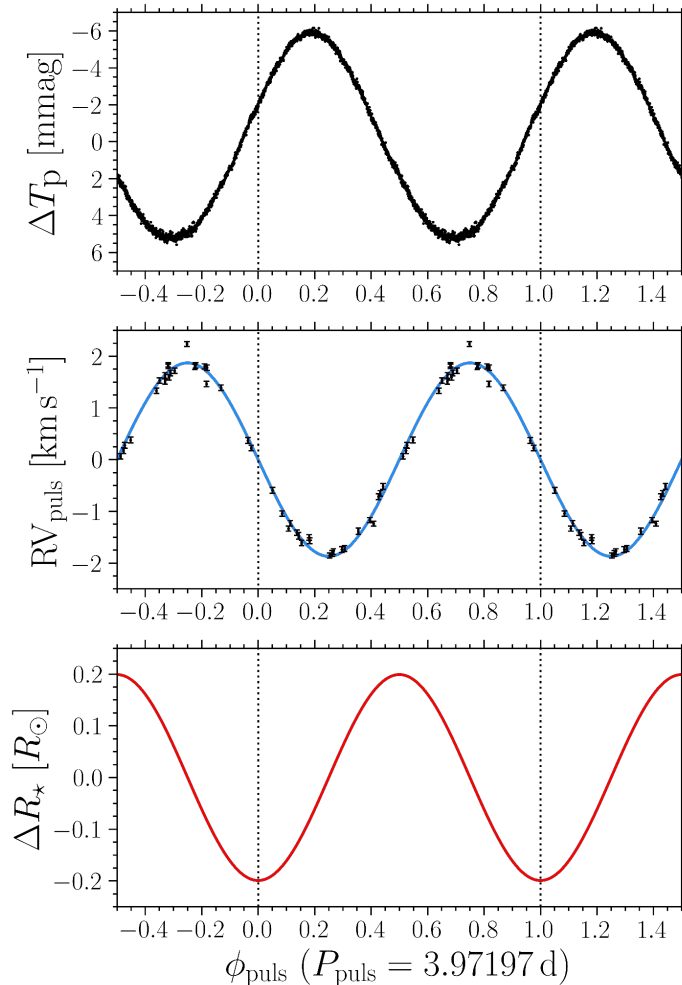
$5.13 \pm 0.28 M_{\odot}$



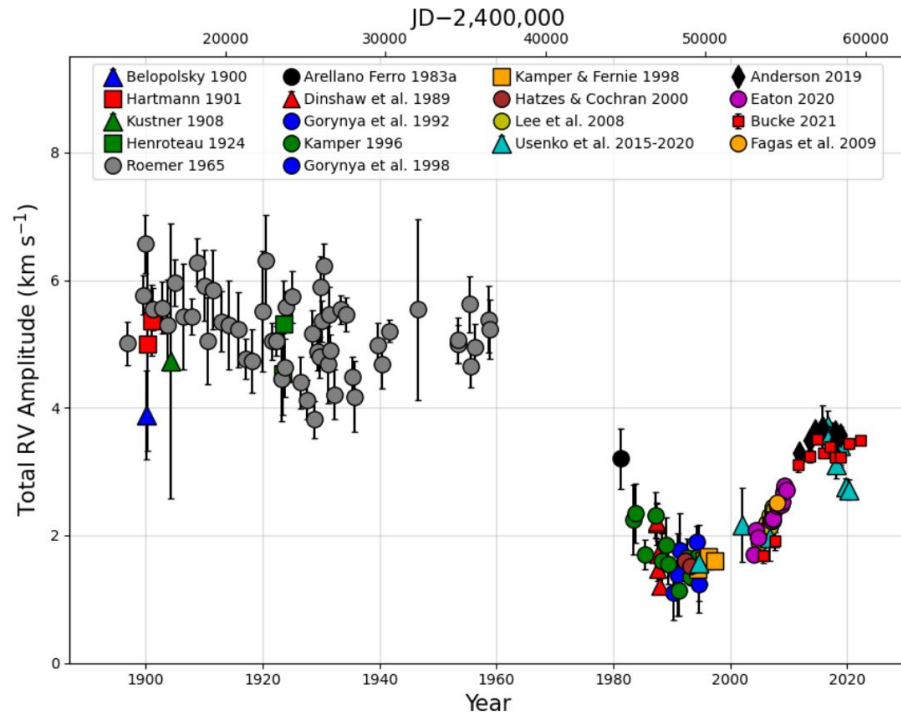
Introduction: Polaris the Cepheid



Low Amplitude Pulsation

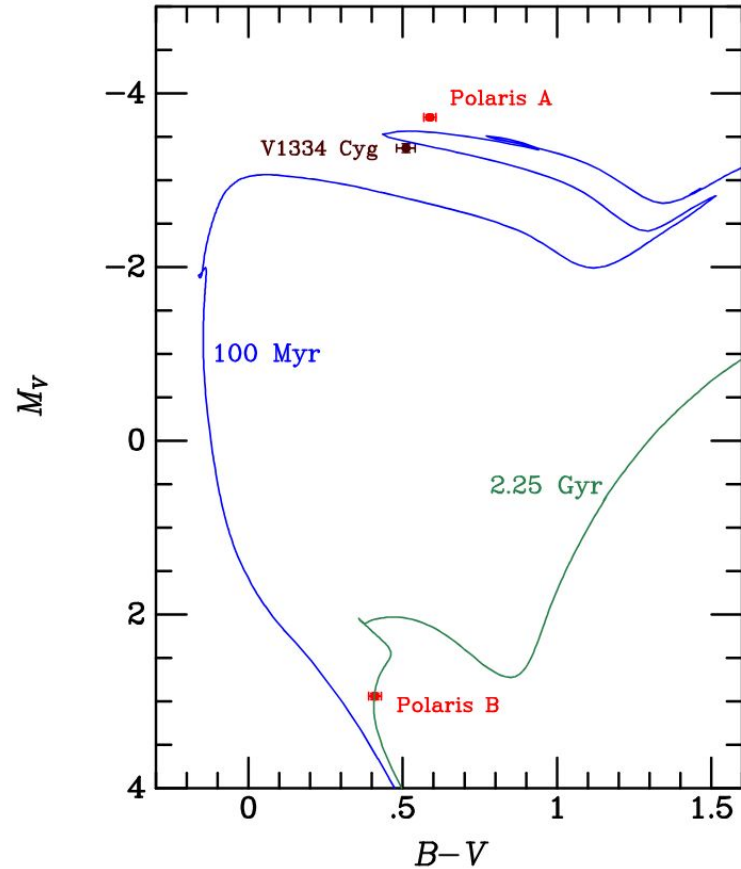


Puzzling Variability Changes



Torres 2021

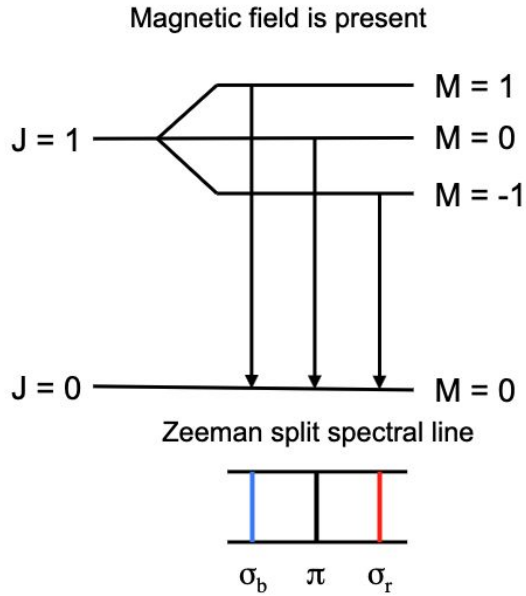
Merger Hypothesis



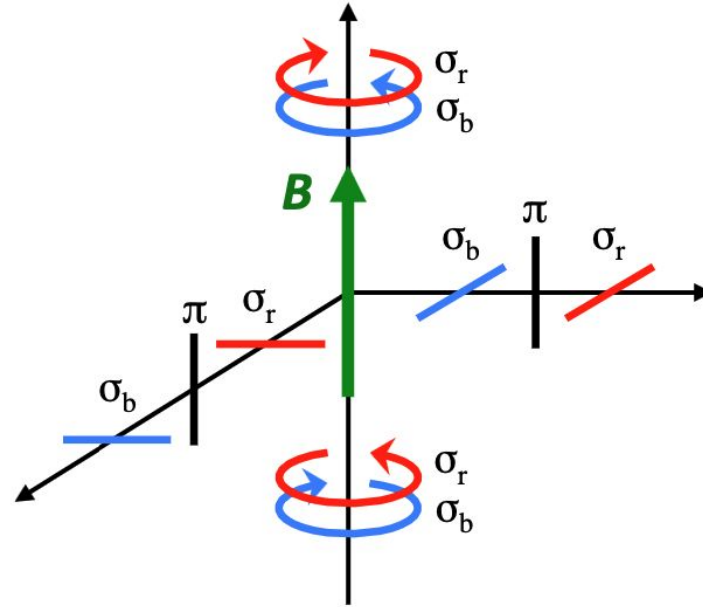
Evans et al. 2018

Introduction: Spectropolarimetry and magnetic fields

Zeeman Line Splitting

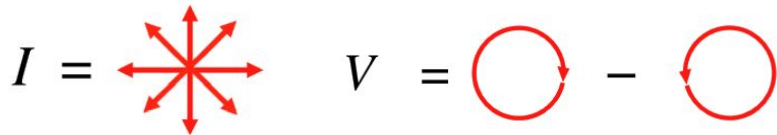
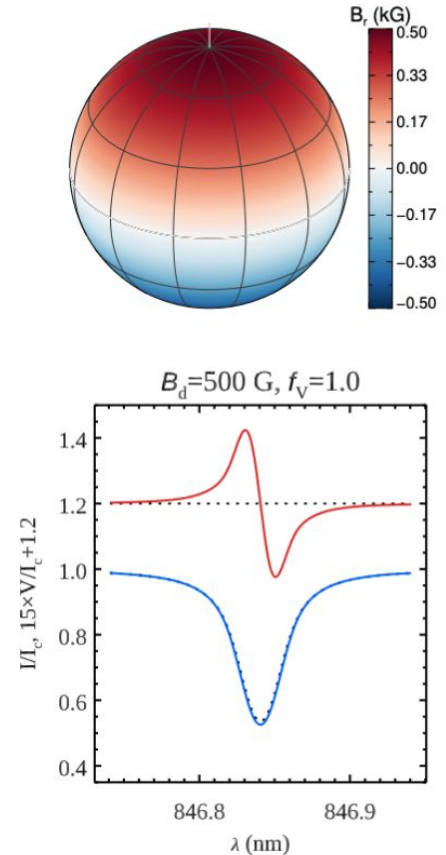


Magnetic Induced Polarization



Kochukhov 2021

Magnetic Field Geometry

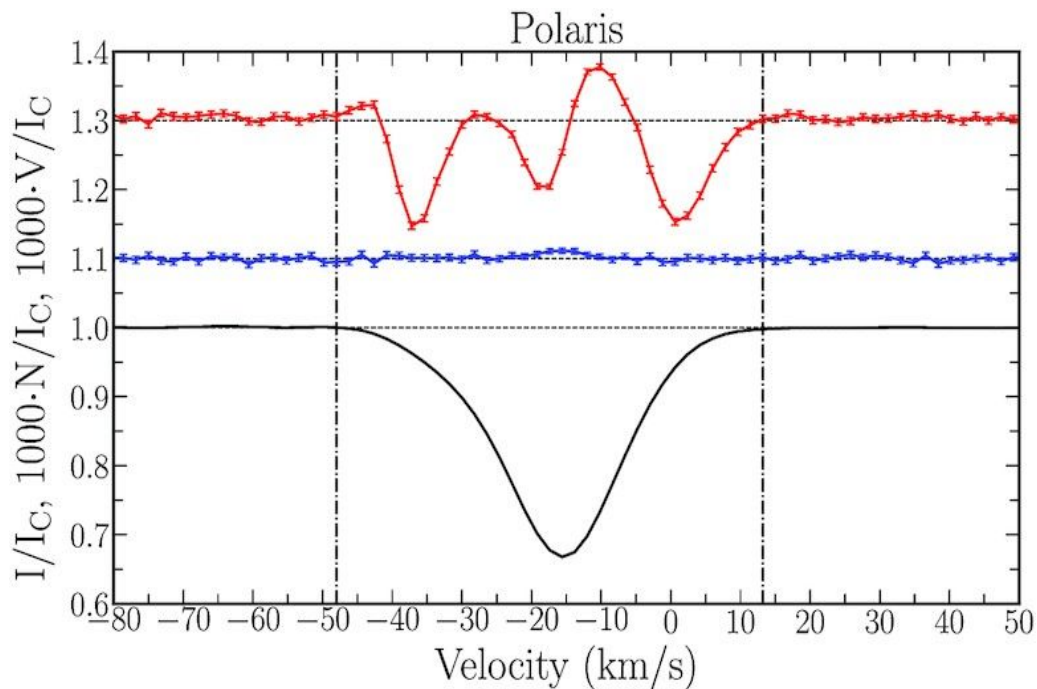


Introduction: Long-term spectropolarimetric monitoring

Project Overview

- Polaris detected in a first magnetic survey of Cepheids (Barron et al. 2022, in prep).
- Initiated a long-term magnetic monitoring campaign with **ESPaDOs** at **CFHT** (2020 - 2025).
- Aimed to measure P_{rot} and study the stability/evolution of the magnetic field.

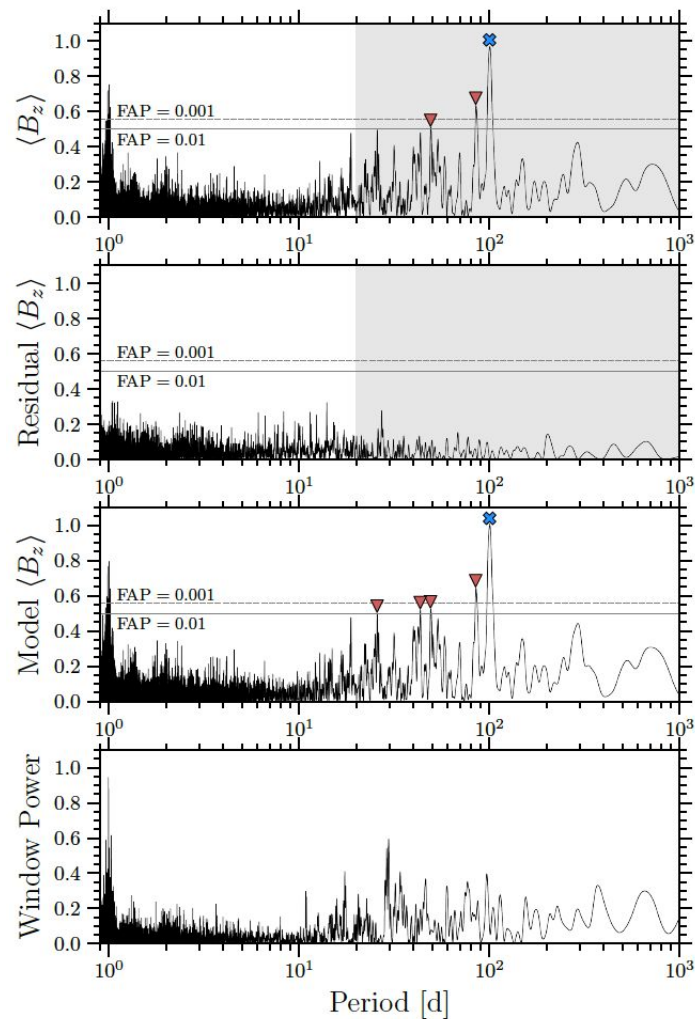
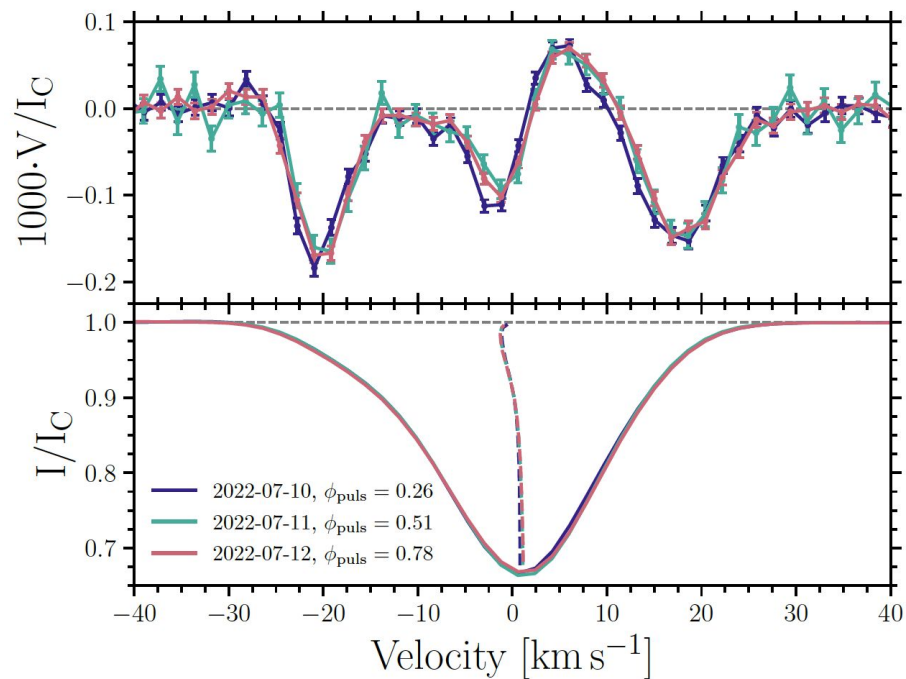
Least-Squares Deconvolution (LSD) Profiles



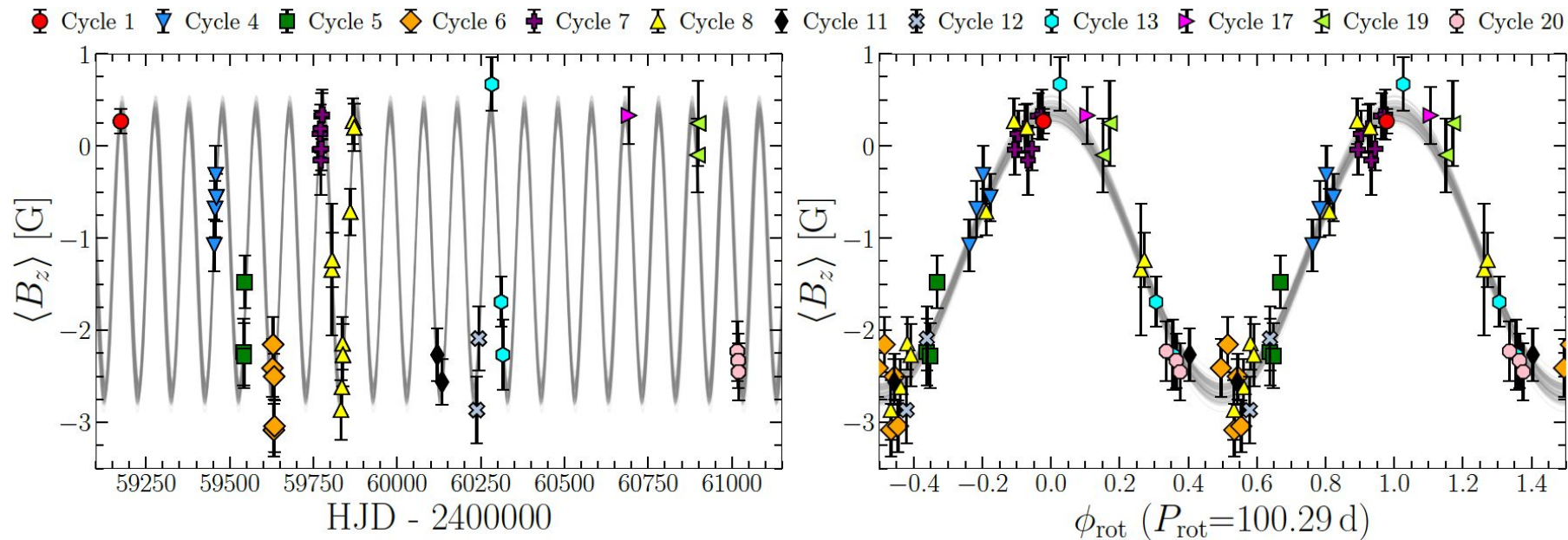
Barron et al. 2022

Results: Period analysis

Magnetic variability not due to pulsation



Results: First direct measurement of P_{rot} for a Cepheid



$$P_{\text{rot}} = 100.29 \pm 0.19 \text{ days}$$

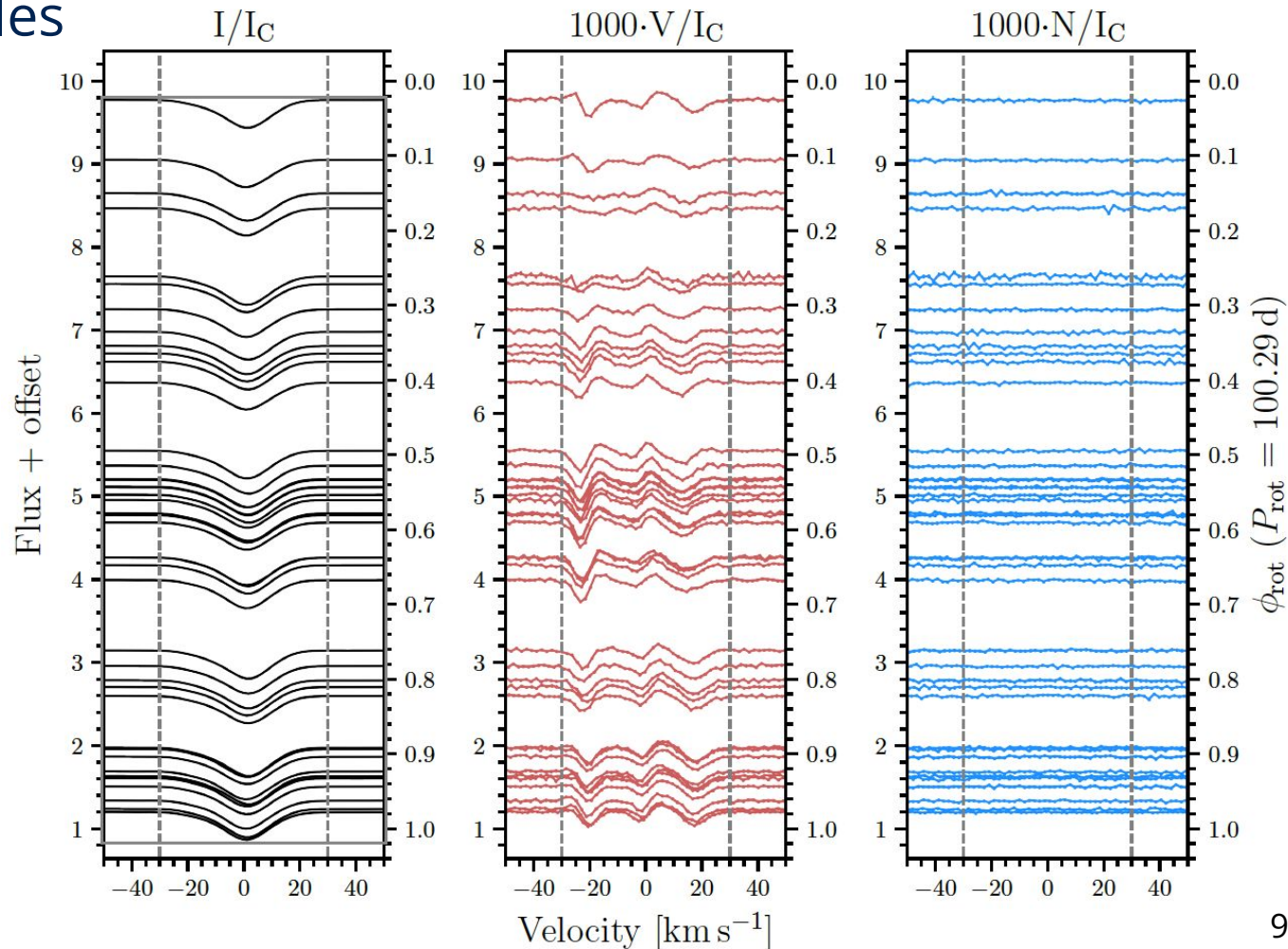
$$v_{\text{eq}} = 23.3 \pm 0.2 \text{ km s}^{-1}$$

$$v_{\text{eq}} = \frac{2\pi \langle R_{\star} \rangle}{P_{\text{rot}}}$$

Results: LSD Profiles

Field Stability

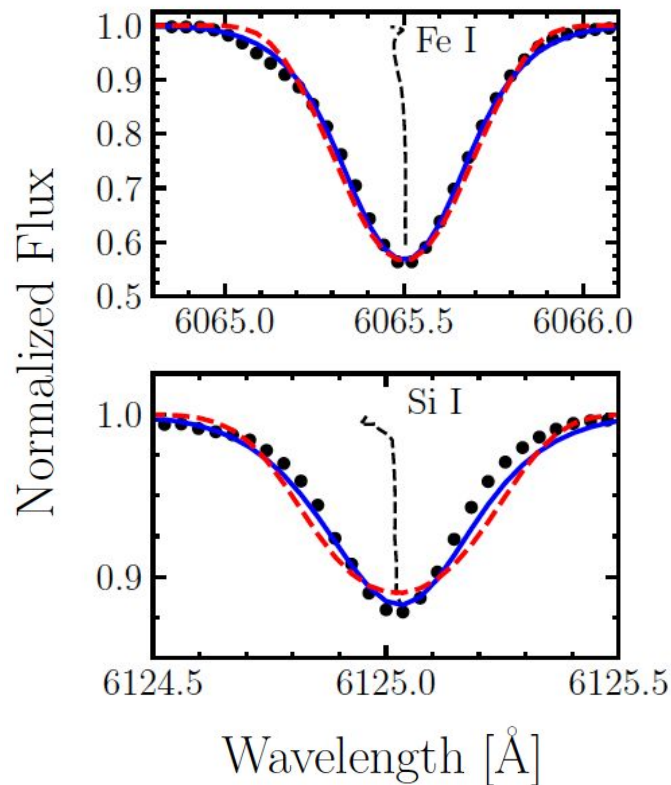
- No clear evidence of magnetic field evolution over 5 years.
- Observations cover 18 rotation and 460 pulsation cycles.



Results: High probability of spin-orbit misalignment

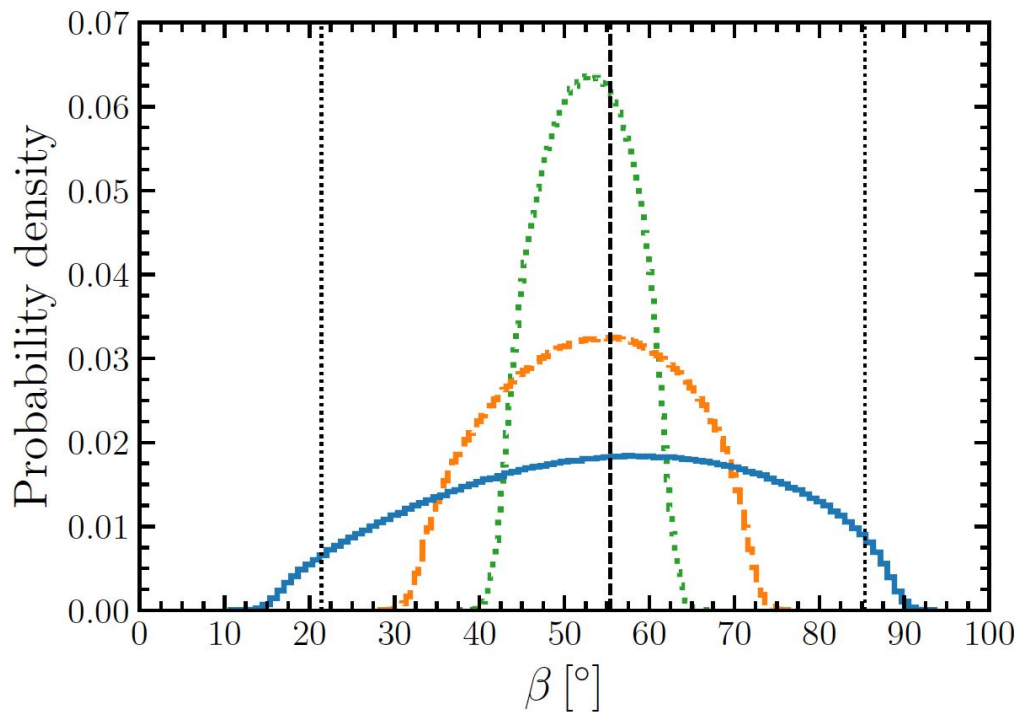
$$v_{\text{eq}} \sin i_{\star} < 13.5 \text{ km s}^{-1} \quad i_{\star} < 37^{\circ}$$

— Best Fit - - - $v_{\text{mac}} = 0 \text{ km s}^{-1}$



Angle between rotation and orbital axes

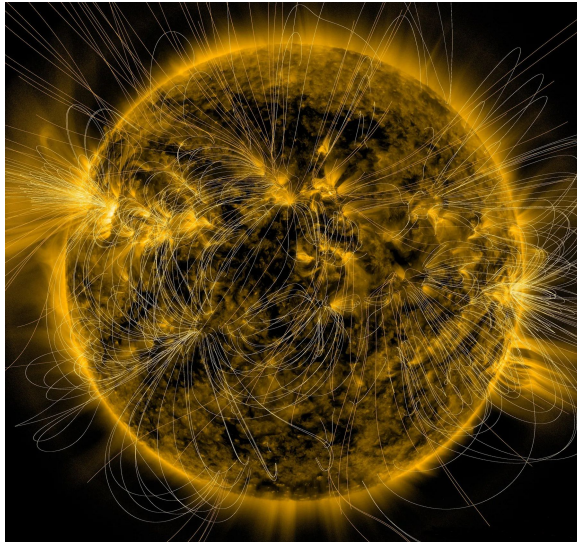
□ $i_{\star} < 37^{\circ}$ □ $i_{\star} < 20^{\circ}$ □ $i_{\star} < 10^{\circ}$



Results: Uncertain origin of the magnetic field

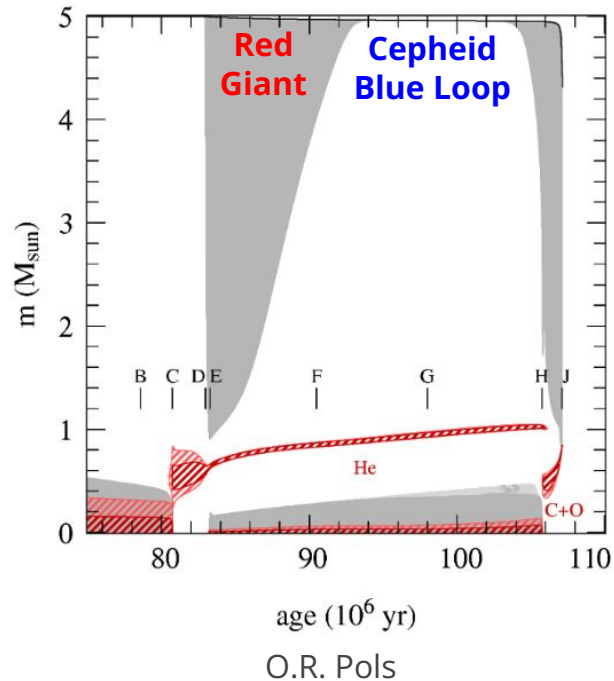
Convective Dynamo?

Solar Dynamo

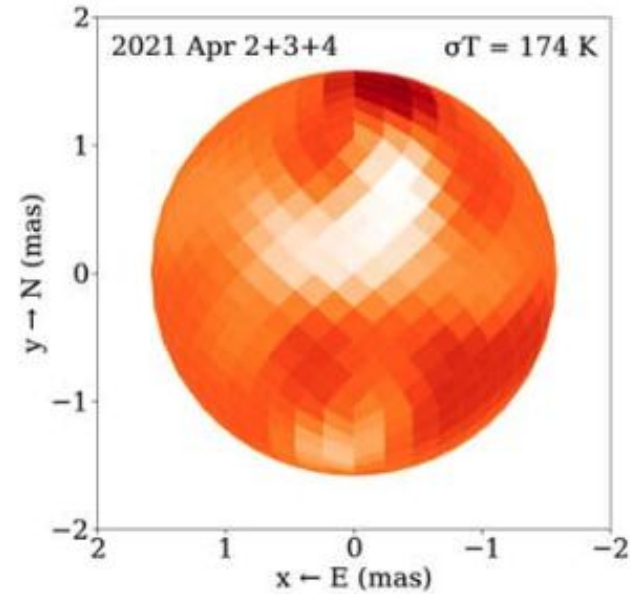


NASA/SDO/AIA/LMSAL

Kippenhahn Diagram



Interferometric Surface Reconstruction

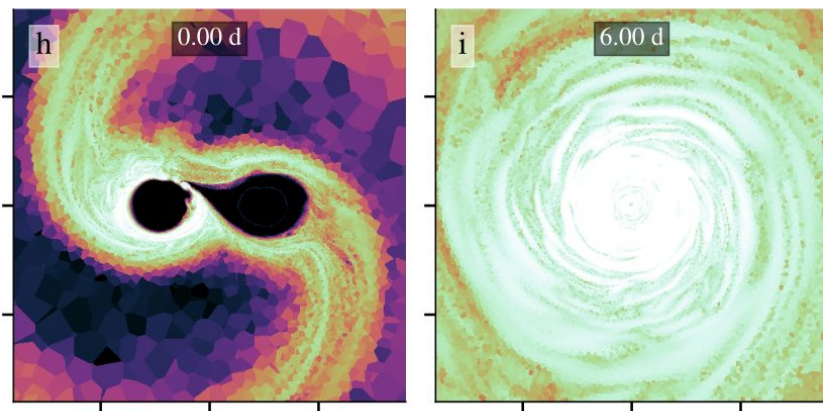


Evans et al. 2024

Results: Uncertain origin of the magnetic field

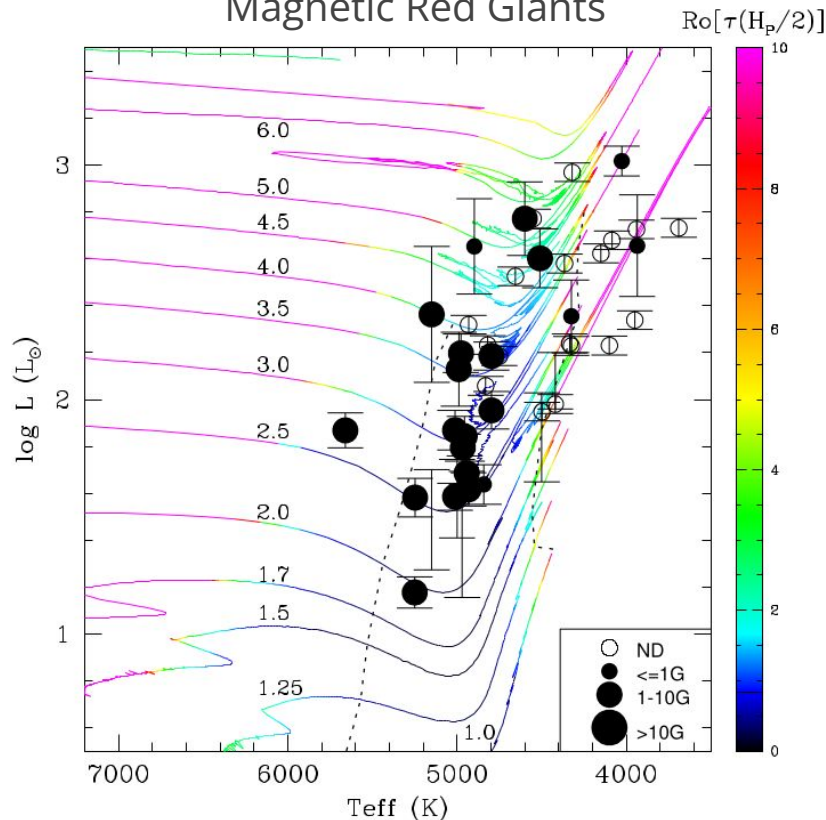
Fossil Field?

Stellar Mergers



Schneider et al. 2019

Magnetic Red Giants



Charbonnel et al. 2017

Conclusions

- Measured a rotation period of ~ 100 days for Polaris, a first for any Cepheid or yellow-supergiant.
- Complexity plus stability of the magnetic field not easily attributed to a dynamo or fossil-field origin.
- Upper limit on rotational broadening implies a high probability of a spin-orbit misalignment.
- Rotational constraints should be incorporated in evolutionary modelling to investigate merger hypothesis.
- Future work should explore reconstruction of Polaris' surface magnetic field geometry.
- Method can be applied to determine rotation periods of blue, yellow and red supergiants.



xkcd.com/2274